

1. Which type of key is a minimal super key with no redundant attributes?
 - A) Super key
 - B) Candidate key
 - C) Foreign key
 - D) Composite key
2. In ER-to-Relational mapping, how is a many-to-many (M:N) relationship typically mapped?
 - A) Add foreign key to either table
 - B) Create a separate junction table
 - C) Merge both tables into one
 - D) Add foreign keys to both tables
3. Which constraint ensures that a foreign key value must reference an existing valid primary key in another table?
 - A) Domain constraint
 - B) Key constraint
 - C) Entity integrity constraint
 - D) Referential integrity constraint
4. What does the wildcard symbol (*) in a SELECT statement do?
 - A) Selects only numeric columns
 - B) Selects all columns from the table
 - C) Selects the first column only
 - D) Selects distinct values
5. Which SQL command is used to remove an entire table along with all its data permanently?
 - A) DELETE TABLE
 - B) REMOVE TABLE
 - C) DROP TABLE
 - D) TRUNCATE TABLE
6. In a 1:N relationship mapping, where should the foreign key be placed?
 - A) On the "one" side table
 - B) On the "many" side table
 - C) In a separate junction table
 - D) In both tables
7. Which constraint prevents duplicate values in a column?
 - A) NOT NULL
 - B) PRIMARY KEY only
 - C) UNIQUE

- D) DEFAULT

8. What does the IS NULL operator do?

- A) Finds records with empty string values
- B) Finds records with zero values
- C) Finds records with missing/no values
- D) Finds records with any value

9. Which of the following is NOT a valid DDL command?

- A) CREATE
- B) INSERT
- C) ALTER
- D) DROP

10. What is the entity integrity constraint?

- A) Foreign key cannot be NULL
- B) Primary key values cannot be NULL
- C) All attributes must have values
- D) No duplicate values anywhere

11. In an ER diagram, how is a weak entity represented?

- A) Single rectangle
- B) Double rectangle
- C) Dashed rectangle
- D) Rounded rectangle

12. Which operation adds new tuples into a relation?

- A) UPDATE
- B) SELECT
- C) INSERT
- D) DELETE

13. What happens when you run DELETE FROM students without a WHERE clause?

- A) Deletes the table structure
- B) Deletes all rows from the table
- C) Deletes the first row only
- D) Nothing, syntax error

14. Which type of attribute can hold multiple values for a single entity instance?

- A) Composite attribute
- B) Derived attribute
- C) Simple attribute
- D) Multivalued attribute

15. The condition "WHERE marks > 80" will return rows where marks is:

- A) 80 or greater
- B) Greater than 80
- C) Less than 80
- D) Equal to 80

16. Which join returns only matching rows from both tables based on a condition?

- A) Outer Join
- B) Natural Join
- C) Inner Join
- D) Cross Join

17. What does the DEFAULT constraint do?

- A) Prevents NULL values
- B) Assigns an automatic value when no value is provided
- C) Creates a primary key automatically
- D) Ensures all values are unique

18. In Chen notation, what shape represents a relationship?

- A) Rectangle
- B) Oval
- C) Diamond
- D) Line

19. What is the correct syntax to add a new column to an existing table?

- A) ALTER TABLE table_name ADD column_name datatype
- B) ADD COLUMN table_name column_name datatype
- C) ALTER TABLE table_name INSERT column_name datatype
- D) MODIFY TABLE table_name ADD column_name datatype

20. Which of the following is a valid comparison operator for "not equal to" in SQL?

- A) !=
- B) <>
- C) Both A and B
- D) NOT =

1. What is the difference between a super key and a candidate key?
2. Explain the difference between DDL and DML commands with one example of each.
3. What is referential integrity? Provide a real-world example.
4. Why are M:N relationships mapped using a separate junction table rather than adding foreign keys to the existing tables?
5. What is the difference between DELETE and DROP commands?
6. Explain the purpose of the WHERE clause in SQL with an example.
7. What is a weak entity? How does it differ from a strong entity?
8. What is the difference between INNER JOIN and NATURAL JOIN?
9. Why must multivalued attributes be placed in separate tables during ER-to-Relational mapping?
10. What is the difference between total participation and partial participation in relationships?

1. **ER-to-Relational Mapping Scenario:** A university database has the following requirements:
 - STUDENT entity with attributes: StudentID (primary key), Name, Email
 - COURSE entity with attributes: CourseID (primary key), Title, Credits
 - INSTRUCTOR entity with attributes: InstructorID (primary key), Name, Department
 - Students enroll in many courses; each course has many students (M:N relationship) with a Grade attribute
 - Each instructor teaches many courses; each course is taught by one instructor (1:N relationship)
 - Each student has one advisor (instructor); each instructor advises many students (1:N relationship)

Perform the complete ER-to-Relational mapping and write the final relational schema with all tables, primary keys, and foreign keys clearly identified.

2. **SQL Query Writing:** Consider the following database tables:

Students (StudentID, Name, City, GPA)

Courses (CourseID, Title, Credits)

Enrollment (StudentID, CourseID, Grade, Semester)

Write SQL queries for each of the following:

- a) Find the names of all students with GPA greater than 3.5
- b) Find the names of students enrolled in 'Database Systems' course
- c) Find all courses (Title and Credits) that have 3 or more credits
- d) Update the grade to 'A' for student with StudentID 101 in CourseID 'CS201'
- e) Find all students whose city is 'Lahore' or 'Islamabad'

3. **Constraints and Update Operations:** Discuss the following scenarios with proper explanations:
- a) You try to insert a new employee record with a DepartmentID that does not exist in the Department table. Which constraint is violated and why?
 - b) You attempt to update a primary key value to a value that already exists in the table. What happens and why?
 - c) Explain the difference between ON DELETE CASCADE, ON DELETE RESTRICT, and ON DELETE SET NULL. When would you use each?
 - d) A student has no email address provided. Should you store NULL or an empty string? Justify your answer.
4. **ER Diagram Construction:** Draw an ER diagram for a Hospital Management System based on the following requirements:
- PATIENT has attributes: PatientID (unique), Name, DOB, Phone
 - DOCTOR has attributes: DoctorID (unique), Name, Specialization, Salary
 - APPOINTMENT has attributes: Date, Time, Status
 - A patient can book many appointments; each appointment is for one patient
 - A doctor can have many appointments; each appointment is with one doctor
 - A patient has one primary doctor; a doctor can be primary doctor for many patients
 - Each appointment has a prescription (multivalued attribute - can have multiple medications)
 - DEPENDENT (weak entity) depends on PATIENT with partial key DependentName
- Identify all entities, attributes, relationships with cardinality ratios, and participation constraints.
5. **Relational Model Concepts:** Explain the following concepts with appropriate examples:
- a) Domain constraints and why they are important for data integrity
 - b) Entity integrity constraint and its purpose
 - c) The difference between a super key, candidate key, and primary key using a STUDENT table with attributes (StudentID, CNIC, Email, Name, Department)
 - d) How the relational model achieves data independence
 - e) What happens when you try to DELETE a parent record that has child records referencing it, and no cascade option is specified? Explain with an example.